

Radial geodesic component

$$\frac{Du^R}{d\tau} = \frac{du^R}{d\tau} + \Gamma^R_{55}(u^5)^2 = 0$$

$$\frac{d^2R}{d\tau^2} = -\Gamma^R_{55}\left(\frac{dy}{d\tau}\right)^2$$

$$\Gamma^R_{55} = -\frac{1}{2}\,\partial_R G_{55}$$

Conceptual correspondence: spatial variation of G_{55} supplies the connection term; the covariant acceleration is zero along the modeled geodesic, while its radial coordinate component appears as the generalized force contribution. The field arrows are illustrative and do not specify a radiation law.